

Problem Set 8: Matrix Algebra, OLS, and Optimization

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Overview

This problem set explores multiple approaches to computing OLS estimates: the closed-form analytical solution, gradient descent, two numerical optimization algorithms (L-BFGS-B and Nelder-Mead), and maximum likelihood estimation. All computations are performed in Python using `numpy`, `scipy`, and `statsmodels`.

Q4: Data Generation

I generate a dataset with $N = 100,000$ observations and $K = 10$ parameters using `numpy.random.seed(100)`. The design matrix X is $N \times K$, where the first column is a vector of ones (intercept) and the remaining nine columns are drawn from $\mathcal{N}(0, 1)$. The error vector $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.5$. The true parameter vector is:

$$\beta = [1.5 \quad -1 \quad -0.25 \quad 0.75 \quad 3.5 \quad -2 \quad 0.5 \quad 1 \quad 1.25 \quad 2]'$$

The outcome is generated as $Y = X\beta + \varepsilon$.

Q5: Closed-Form OLS

The OLS estimator is computed analytically as:

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'Y$$

using `numpy.linalg.solve` for numerical stability. The estimates are shown in Table 2. The maximum absolute deviation from the true β is 0.0023, which is very small given that sampling variability is expected at finite N .

Q6: Gradient Descent

The OLS loss function is $\mathcal{L}(\beta) = \frac{1}{2}\|Y - X\beta\|^2$, whose gradient is:

$$\nabla_{\beta}\mathcal{L} = -X'(Y - X\beta)$$

Starting from $\beta^{(0)} = \mathbf{0}$, I iterate the update rule:

$$\beta^{(t+1)} = \beta^{(t)} - \alpha \cdot \nabla_{\beta} \mathcal{L}(\beta^{(t)})$$

with learning rate $\alpha = 3 \times 10^{-7}$ for 10,000 iterations. The algorithm converges to essentially the same solution as the closed-form estimator (max absolute difference: 0.0023).

Q7: L-BFGS-B and Nelder-Mead

I minimize the OLS objective using two algorithms from `scipy.optimize.minimize`:

- **L-BFGS-B**: A quasi-Newton method that uses gradient information. It converges quickly and recovers the closed-form OLS estimates exactly.
- **Nelder-Mead**: A derivative-free simplex method. Despite not using gradients, it also converges to the same solution (max absolute difference: 0.0023).

Both methods yield identical estimates, confirming that the OLS problem has a unique global minimum. L-BFGS-B is substantially faster due to its use of gradient information; Nelder-Mead is slower but demonstrates that gradient-free methods can still be effective here.

Q8: MLE via L-BFGS-B

Assuming $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, the log-likelihood is:

$$\ell(\beta, \sigma) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|Y - X\beta\|^2$$

I minimize the negative log-likelihood over $\theta = (\beta', \sigma)'$ using L-BFGS-B with the gradient:

$$\frac{\partial(-\ell)}{\partial\beta} = -\frac{X'(Y - X\beta)}{\sigma^2}, \quad \frac{\partial(-\ell)}{\partial\sigma} = \frac{N}{\sigma} - \frac{\|Y - X\beta\|^2}{\sigma^3}$$

The MLE estimates of β are identical to OLS (as theory predicts under normality), and $\hat{\sigma}_{\text{MLE}} = 0.4991$, very close to the true $\sigma = 0.5$.

Q9: statsmodels OLS and Coefficient Comparison

I use `statsmodels.api.OLS` with no constant added (the intercept is already the first column of X), equivalent to `lm(Y ~ X - 1)` in R. The regression output is shown below.

Table 2 compares all estimation methods side by side.

Table 1: statsmodels OLS Regression Results

Variable	Coef.	Std. Err.	<i>t</i> -stat	<i>p</i> -value
β_1 (const)	1.4991	0.002	949.72	0.000
β_2	-0.9991	0.002	-630.61	0.000
β_3	-0.2494	0.002	-157.55	0.000
β_4	0.7490	0.002	473.94	0.000
β_5	3.4992	0.002	2212.52	0.000
β_6	-1.9984	0.002	-1267.38	0.000
β_7	0.4992	0.002	315.24	0.000
β_8	0.9977	0.002	631.12	0.000
β_9	1.2520	0.002	791.96	0.000
β_{10}	1.9977	0.002	1264.03	0.000
N	100,000			
R^2	0.990			

Table 2: Comparison of $\hat{\beta}$ Across Estimation Methods

	True β	Closed-Form	Gradient Desc.	L-BFGS-B	Nelder-Mead	MLE
β_1	1.5000	1.4991	1.4991	1.4991	1.4991	1.4991
β_2	-1.0000	-0.9991	-0.9991	-0.9991	-0.9991	-0.9991
β_3	-0.2500	-0.2494	-0.2494	-0.2494	-0.2494	-0.2494
β_4	0.7500	0.7490	0.7490	0.7490	0.7490	0.7490
β_5	3.5000	3.4992	3.4992	3.4992	3.4992	3.4992
β_6	-2.0000	-1.9984	-1.9984	-1.9984	-1.9984	-1.9984
β_7	0.5000	0.4992	0.4992	0.4992	0.4992	0.4992
β_8	1.0000	0.9977	0.9977	0.9977	0.9977	0.9977
β_9	1.2500	1.2520	1.2520	1.2520	1.2520	1.2520
β_{10}	2.0000	1.9977	1.9977	1.9977	1.9977	1.9977

How Close Are the Estimates to Ground Truth?

All five methods produce virtually identical estimates and recover the true β with high accuracy. The maximum absolute deviation across all coefficients and all methods is just **0.0023**, and all deviations are well within the 95% confidence intervals reported by `statsmodels` (standard errors ≈ 0.002). This is expected: with $N = 100,000$ observations, OLS is highly precise, and any deviation from the true β reflects sampling variability from ε rather than estimation error.

Three key observations emerge:

1. The closed-form solution, gradient descent, both optimization algorithms, and MLE all converge to the same $\hat{\beta}$, confirming that the OLS problem has a unique global minimum.
2. The MLE estimate of σ ($\hat{\sigma} = 0.4991$) is very close to the true value $\sigma = 0.5$, validating

the data-generating process.

3. The $R^2 = 0.990$ indicates that the linear model explains nearly all of the variation in Y , consistent with a signal-to-noise ratio of $\text{Var}(X\beta)/\text{Var}(\varepsilon) = \text{large}/0.25$.